

Supplementary data

Can current surveillance follow doxy-PEP-associated *Staphylococcus aureus* resistance?
A multi-stream evidence audit

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All supplementary material is relocated from the validated analysis and regenerates from the public repository (`make all; outputs/`), archived with a citable DOI. US spelling is used (at author request); *JAC* nomenclature conventions are otherwise followed and MICs are expressed in mg/L.

S1. The measurement-inheritance schema

We make the construct operational so it can be applied beyond this case. Treat the answerability of a question as requiring a measurement vector

$$M = \{U, E, P, D, T, L, R\}$$

where *U* is the unit/population, *E* the exposure, *P* the phenotype/outcome, *D* the denominator, *T* the temporal structure/cadence, *L* the linkage at which *E* and *P* are joined, and *R* the locus of measurement responsibility. Measurement inheritance occurs when a downstream system retains the dimensions appropriate to the upstream question but not those the downstream estimand requires. A distributive corollary follows when the benefit and the potential downstream harm accrue to different populations ($B \neq H$): the beneficiary population *B* remains represented by the inherited $\{U, E, P, D, L\}$, while the population *H* that would bear an externalised consequence is represented by none of them jointly. The three streams instantiate the schema directly (Table 1).

Table 1: The measurement vector across the evidence chain. “Present” = the dimension exists at the resolution the downstream question needs; “wrong unit/resolution” = present but inherited from the upstream question; “absent” = not captured.

Dimension	Stream A (trial)	Stream B (guidance)	Stream C (surveillance)
<i>U</i> unit	Participant <i>S. aureus</i>	Not specified for <i>S. aureus</i>	MRSA subset (wrong unit)
<i>E</i> exposure	Randomized doxy-PEP	Prescribing context	PrEP proxy only
<i>P</i> phenotype	Incident doxy-R (present)	Not measured (absent)	Methicillin axis (wrong unit)
<i>D</i> denominator	Randomized participants	None for <i>S. aureus</i>	Population, but methicillin
<i>T</i> cadence	Time-to-event	None	Cross-sectional/facility
<i>L</i> linkage	Individual	None assigned	No common <i>E–P</i> denominator (absent)
<i>R</i> responsibility	Trial investigators	Unassigned (absent)	Unassigned for tetracycline <i>S. aureus</i>

The two load-bearing gaps are *R* in Stream B (no party assigned to measure the bystander) and *L* in Stream C (no deployed system joins doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator).

S2. Stream A — supplementary trial analyses

These analyses establish that the cross-sectional carriage endpoint could not resolve the downstream population question under the observed designs and sampling structure. They are MRSA-scoped (MRSA being a subset, not the analytic unit) and are *not* load-bearing for the individual-level result the final trial resolved.

- **Detectability (interim).** For each first-party primary-trial *S. aureus* arm comparison, exact-Fisher power and the minimum detectable RR at 80% power were computed from arm sizes and control rate against the external benchmark. None of four comparisons was powered; the minimum detectable RR was 3.8–5.1 against a benchmark of 1.42–2.25 (`outputs/detectability_result.md`).
- **Duration/accrual.** Preserving each comparison’s arm ratio and control rate, the sample reaching 80% power lands in the hundreds to thousands per arm; the realised denominator reflects the early efficacy stop (`outputs/duration_result.md`). No duration counterfactual is inferred from the two arm-split visits.
- **Clustering dispersion.** Between-cohort dispersion of MRSA colonization across independent MSM cohorts was estimated by a logit-normal binomial random-effects model (Gauss–Hermite quadrature) as a conservative external heterogeneity anchor (`outputs/dejong_sigma_result.md`).
- **Trend-test degradation.** At the trials’ per-visit denominators, a naive cross-sectional trend test (Cochran–Armitage) is badly oversized under the estimated dispersion, scoring a flat process as a significant trend most of the time (`outputs/coverage_null_result.md`).
- **Outbreak-shape non-identifiability.** Flat, monotone-trend and outbreak-wave models fitted to the MRSA carriage series by maximum likelihood are not distinguishable by AIC or parametric-bootstrap likelihood-ratio tests (`outputs/three_outbreak_fit_result.md`).
- **Mechanism-blind measurement in *S. aureus* (now in the main text).** In a military antimalarial–doxycycline cohort of 168 personnel, overall tetracycline-class resistance in *S. aureus* did not differ by doxycycline exposure while *tet*(M) genes were significantly more common among the exposed ($p=0.031$) — a single breakpoint concealing a mechanism shift. This cohort is *S. aureus*-specific (not cross-organism); it is promoted to the main-text phenotype-unit argument (Mende *et al.*, *Diagn Microbiol Infect Dis* 2016;86:211–220).

The one positive within-arm trend statistic in this material is a published-figure re-analysis of DOXYVAC MRSA carriage (1.8% to 9.9%), cited as such and not originated here.

S3. Stream B — guidance corpus and coding

Denominator. The purposive denominator is the US jurisdictions with documented community-associated MRSA/USA300 transmission in men who have sex with men, defined by de Jong’s systematic review and the outbreak literature it cites — the same source supplying the Stream A clustering dispersion. Governmental public-health guidance only was assessed; clinic and provider-organization documents were excluded because they carry neither surveillance authority nor monitoring policy.

Coding. Each document was coded from full text (not search snippets), one record per document, every finding carrying a locator, for: whether it mandates in-category monitoring (gonorrhoea, chlamydia, syphilis) and host-toxicity labs; and whether it names, counsels on, or

measures *S. aureus*. Bystander treatment was graded silent / generic-microbiome / organism-named; *S. aureus* monitoring as required / suggested / silent / explicitly-none (the last requiring a verbatim quote). Coding was blind to the expected outcome. A blinded second coder double-coded with per-field Cohen’s κ ; agreement was 92% and the single field with meaningful disagreement ($\kappa = 0.58$) is reported unrounded.

Result. Of the six documented-outbreak jurisdictions — San Francisco, Chicago, New York City, Boston (Massachusetts), Los Angeles County and San Diego — 0 require or suggest *S. aureus* monitoring; three name *S. aureus*, two invoke only a generic microbiome risk, one is silent. An independence check from document content (structure, shared passages, stated derivation) finds four independently authored declines, with Los Angeles County adapted from San Francisco and Massachusetts from an external template. Across the full coded corpus (10 documents: CDC, WHO and eight US jurisdictions) 0 instrument the bystander; a further jurisdiction deferring to CDC inherits CDC’s non-monitoring and is counted separately. The full per-document coding table with locators is in the repository (data/processed/).

S4. Stream C — surveillance enumeration and dilution model

Enumeration. A closed universe of twelve established US governmental systems (federal, state or metropolitan) that could plausibly capture, at a population denominator, either doxy-PEP/doxycycline exposure or *S. aureus* AMR phenotype was coded on three capabilities: exposure capture, phenotype capture (of the *S. aureus* tetracycline unit, not the MRSA subset) and the join. Two capture exposure at a population denominator (AIDSVu; CDC AtlasPlus) as a PrEP proxy; zero capture *S. aureus* tetracycline resistance at a population denominator. The population-based *S. aureus* platform (ABCs/EIP) and the National Healthcare Safety Network report the methicillin axis; the facility cumulative antibiogram is the only routine tetracycline reading and carries no population denominator (CLSI M39 mandates the MRSA/MSSA split and makes tetracycline a selectively suppressible supplemental agent); the National Antimicrobial Resistance Monitoring System excludes *S. aureus*. Tetracycline-resistant *S. aureus* is nationally notifiable nowhere. The full system-by-system matrix with geographic resolution is in the repository.

Dilution model. The exposed-subgroup dilution fraction is

$$f = \phi(\text{male-PrEP rate}/10^5) \text{ uptake } \kappa,$$

population-independent because it tracks exposure density, not headcount. The minimum detectable change in a proportion near baseline R_0 (two-sided $\alpha = 0.05$, 80% power) inverts to $RR_{\text{needed}} = 1 + \text{MDE}/(fR_0)$. Parameters were bounded to favor detection, so a failed gate is robust: uptake $\in \{20, 35, 55\}\%$; $\kappa \in \{0.2, 0.5, 1, 3, 5\}$ ($\kappa < 1$ is the realistic region, as the exposed are under-represented in a population isolate stream); $R_0 \in \{5, 10, 13\}\%$; isolate volume $N \in \{1\,000, 10\,000, 100\,000\}$ per geography-year (100 000 exceeds any real US stream). Under the realistic cell, $RR_{\text{needed}} \approx 14\text{--}64$ across the design-effect range; detection appears only in a best-case cell ($RR \approx 1.93$) that compounds four simultaneously implausible assumptions and fails on any one alone. The metropolitan break-even uses Washington, D.C. (2 694 per 100 000 adult males) as the observed density ceiling and requires 2.1–33 times that density, or roughly 325% of adult males on PrEP under realistic isolate volumes (Figures A–C).

Alt text: A heatmap grid of doxy-PEP uptake against isolate enrichment in which every cell shows a required relative risk above 1.42.

Alt text: A chart showing required male-PrEP density in multiples of the densest US metropolitan geography, with all requirements above the one-times national maximum.

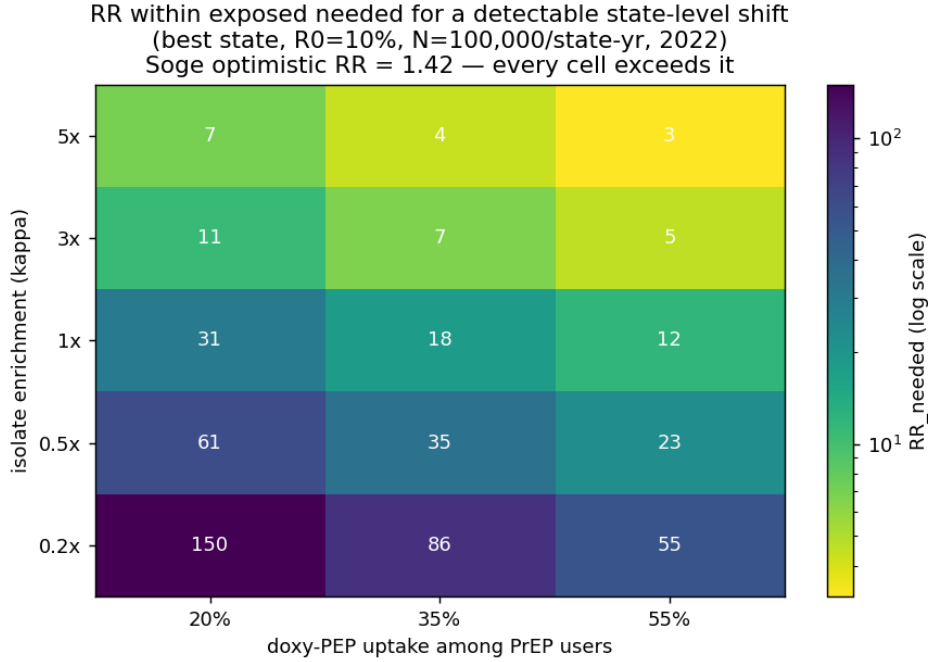


Figure 1: *

Figure A. State level. Within-exposed RR required for a detectable state-level shift in tetracycline non-susceptibility, over doxy-PEP uptake $\times$ isolate enrichment (best state, $R_0 = 10\%$, $N = 100\,000/\text{state-year}$). Every cell exceeds the optimistic benchmark $RR = 1.42$.

Alt text: Break-even curves of the share of all adult males on PrEP required for detection against isolate volume, all lying above both the observed 2.7% and the 100% physical ceiling.

S5. Literature attention structure

A defined PubMed search (2015–2026; queries and a frozen result snapshot in the repository) identified 361 doxy-PEP papers, of which 13 (3.6%) mention a bystander staphylococcal organism and only 2 measure tetracycline-resistant *S. aureus* against a doxy-PEP exposure contrast. This is consistent with a research system whose attention remains concentrated where its existing instruments already point, and is not evidence of neglect by any author.

S6. Distributive observability

The three streams share a measurement asymmetry that is a property of the evidence architecture before it is a moral claim. The intervention’s benefit and the trial’s resistance signal are identifiable within an exposed population: exposure ascertained, denominator defined, phenotype measured, a longitudinal causal contrast available. The population that would inherit a downstream resistance consequence has none of these jointly — no assigned measurement mandate (Stream B) and no deployed system in which doxy-PEP exposure and the *S. aureus* tetracycline phenotype share a denominator (Stream C). We call this difference in scientific visibility *distributive observability*: which population remains represented by the evidence system, and at what inferential resolution, once the question moves downstream from the treated to the affected population. It does not establish that a population harm has occurred; it establishes that, were one occurring, the deployed architecture could estimate the intervention side of the benefit–externality ledger far more completely than the externality side. An externality no in-

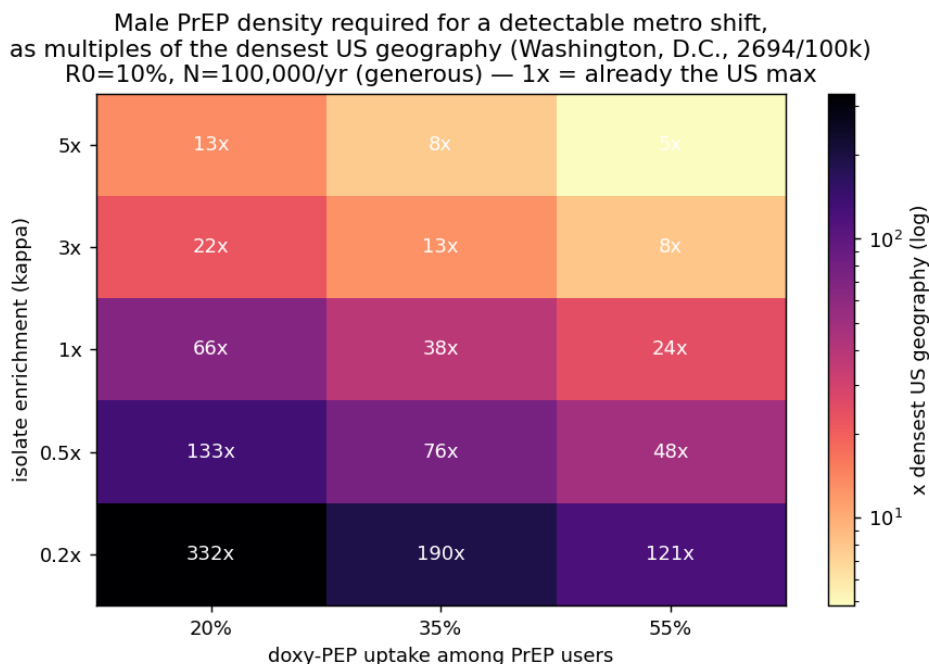


Figure 2: *

Figure B. Metro level. Male-PrEP density required for a detectable shift, as multiples of the densest US geography (Washington, D.C.). 1× is already the national maximum.

strument can price is not thereby small — it is unweighable, and an architecture that prices only one side of the ledger cannot show that benefit outweighs external cost. The construct is built from familiar parts (estimand mismatch; ascertainment and surveillance bias; metric substitution of the Goodhart type) but concerns their coupling across an evidence-production chain, and generalises to any intervention whose benefits concentrate in an observed recipient population while its consequences diffuse beyond it.

S7. Reproducibility

All analysis is Python 3.11 (pandas, statsmodels, scipy, matplotlib); no proprietary dependencies. Raw data are immutable; all cleaning is in code; every figure and table regenerates from `make all`. Analysis decisions that could go either way were logged with dates in `DECISIONS.md` before results were known. All source files are SHA-256-pinned (`data/raw/**/CHECKSUMS.md`). The repository is public and archived with a citable DOI.

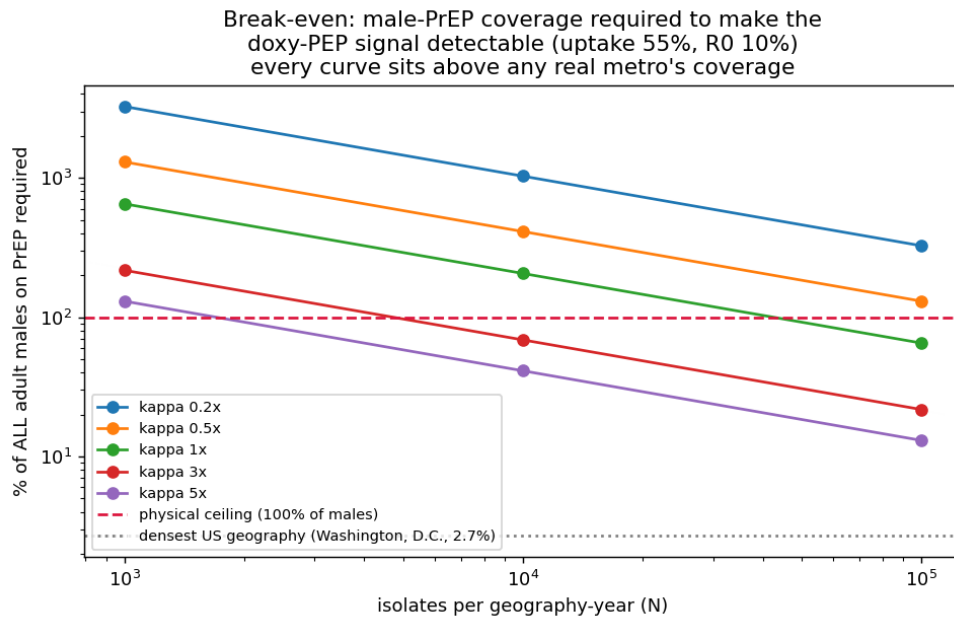


Figure 3: *

Figure C. Break-even in proxy-free units. Share of *all* adult males on PrEP required for detection versus isolates per geography-year, by enrichment κ . Every curve sits above D.C.'s observed 2.7% and, under realistic isolate volumes, above the 100%-of-males physical ceiling.